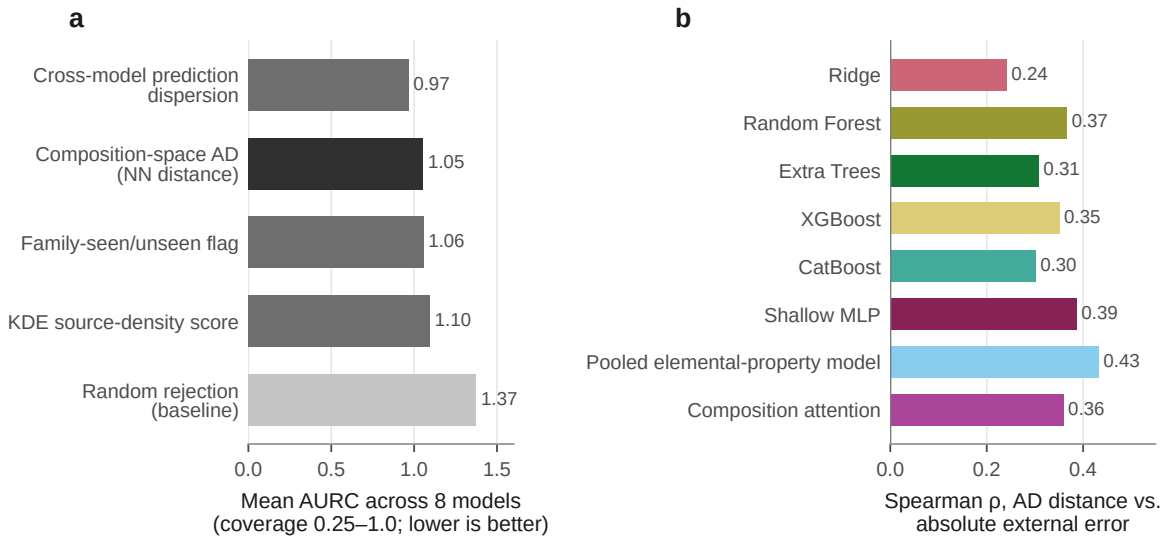
a
Cross-model prediction dispersion
0.97
Composition-space AD (NN distance)
1.05
Family-seen/unseen flag
1.06
KDE source-density score
1.10
Random rejection (baseline)
1.37
0.0
0.5
1.0
1.5
Mean AURC across 8 models
(coverage 0.25–1.0; lower is better)
b
Ridge
0.24
Random Forest
0.37
Extra Trees
0.31
XGBoost
0.35
CatBoost
0.30
Shallow MLP
0.39
Pooled elemental-property model
0.43
Composition attention
0.36
0.0
0.2
0.4
Spearman ρ, AD distance vs.
absolute external error